Seed-cum-Fertilizer Drill



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 5 labs

Covers these lab items: Seed cum fertilizer; Seed-cum-Fertilizer Drill

What it is used for

A seed-cum-fertilizer drill places seeds and fertilizer in rows during a single field operation. Correct calibration, suitable placement, and careful operation help achieve uniform sowing while reducing seed waste and uneven fertilizer distribution.

Main parts

- Seed hopper stores seed for sowing.
- Fertilizer hopper stores fertilizer for application.
- Metering mechanisms regulate seed and fertilizer delivery.
- Seed tubes guide seed from the metering unit to the furrow.
- Fertilizer tubes direct fertilizer to its intended placement point.
- Furrow openers create channels in the soil.
- Drive wheels and transmission operate the metering mechanisms.
- Frame, hitch, and depth adjustment components support the implement.

Before you start

1. Read the manual and identify suitable seed, fertilizer, and field settings.
2. Inspect the frame, hitch, wheels, chains, tubes, openers, and guards for damage.
3. Confirm the tractor or power unit is compatible with the drill.
4. Calibrate seed and fertilizer rates according to the recommended procedure.
5. Check that seed and fertilizer tubes are clear and correctly connected.
6. Wear suitable footwear and PPE, and remove people from the implement's working area.

How to use

1. Park the tractor and implement safely, lower the drill, and stop the power source before adjustments.
2. Set the row spacing and sowing depth according to the crop requirements.
3. Fill the seed and fertilizer hoppers with the correct materials without exceeding capacity.
4. Connect and secure the implement to the compatible tractor hitch.
5. Raise the implement for transport and lower it into working position at the field edge.
6. Engage the drive mechanism according to the manual and begin moving at a suitable speed.
7. Monitor seed placement, fertilizer delivery, furrow coverage, and row spacing during sowing.
8. Stop in a safe location and inspect the sowing pattern if irregular delivery occurs.
9. Make adjustments only after stopping the machine and isolating relevant power sources.

10. At the end, empty remaining materials as appropriate, clean the tubes and hoppers, and inspect the implement.

⚠ Safety

- Keep hands and feet away from metering mechanisms, chains, drive shafts, and furrow openers.
- Never clear blocked tubes while the drill or tractor power take-off is moving.
- Stop the tractor, lower the implement, apply the parking brake, and isolate power before servicing.
- Handle fertilizers according to the product label and avoid skin, eye, and inhalation exposure.
- Keep bystanders away during turning, reversing, hitching, and field operation.
- Use guards and never ride on the drill unless the manufacturer specifically provides an approved operator position.

Care & storage

- Remove seed and fertilizer residues after use to reduce corrosion and cross-contamination.
- Inspect seed tubes, fertilizer tubes, openers, chains, and metering parts for wear or blockage.
- Lubricate approved points and maintain drive components according to the manual.
- Store the implement in a dry area with the unit safely supported.

Common mistakes to avoid

- Skipping calibration and using an unsuitable seed or fertilizer rate.
- Setting the sowing depth incorrectly for the crop and soil conditions.
- Allowing seed and fertilizer tubes to remain blocked or incorrectly connected.
- Adjusting or clearing the drill while drive components are moving.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.